CSA1708-ARTIFICIAL INTELLIGENCE

LAB EXPERIMENTS

Program 11: WRITE THE PYTHON POGRAM FOR MAP COLORING TO IMPLEMENT CSP

Aim:

To write a Python program to implement the Map Coloring problem using a Constraint Satisfaction Problem (CSP), ensuring that no two adjacent regions have the same color.

Algorithm:

1. Start the program.
2. Define the map regions and their neighboring regions.
3. Define the available colors.
4. Initialize an empty dictionary to store the color assignments.
5. Select an uncolored region.
6. Assign a color that does not conflict with any adjacent region.
7. If no valid color is available, backtrack and try another color.
8. Repeat until all regions are colored successfully.
9. Display the color assigned to each region.
10. Stop the program.

PY8.py - C:/Users/navit/PY8.py (3.12.10)

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```
def is_valid(region, color, assignment, graph):
    for neighbor in graph[region]:
        if assignment.get(neighbor) == color:
            return False
    return True


def map_coloring(graph, colors, assignment):
    if len(assignment) == len(graph):
        return True

    region = next(r for r in graph if r not in assignment)

    for color in colors:
        if is_valid(region, color, assignment, graph):
            assignment[region] = color

            if map_coloring(graph, colors, assignment):
                return True

            del assignment[region]

    return False


graph = {
    'A': ['B', 'C'],
    'B': ['A', 'C', 'D'],
    'C': ['A', 'B', 'D'],
    'D': ['B', 'C']
}

colors = ['Red', 'Green', 'Blue']
assignment = {}

if map_coloring(graph, colors, assignment):
    print("Map Coloring Solution:")
    for region, color in assignment.items():
        print(region, ":", color)
else:
    print("No solution exists")
```

IDLE Shell 3.12.10

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```
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr  8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========================= RESTART: C:/Users/navit/PY8.py =========================
Map Coloring Solution:
A : Red
B : Green
C : Blue
D : Red
>>>
```

Result:

The Python program was executed successfully, and the Map Coloring problem was solved using the CSP backtracking algorithm.